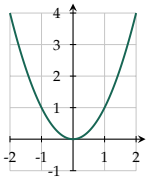
# 8 Variations and extrema of a function

**Sheet 43** Thu 11 Feb. 2027 *Increasing, decreasing, monotonicity*

1. Warm-up · Quick $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; what do you notice?
2. Core · MCQ $f$ is increasing on $I$ means that, for all $a < b$ in $I$:
   **A.** $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$
3. Core + · Graph Curve of $f(x) = x^2$ on $[-2, 2]$:




   Give the variations of $f$.
4. Review · Quick Which multiplier corresponds to an increase of 15%? To a decrease of 15%?
5. Thread · Quick Is 457 divisible by 3? By 9? Justify.

**Sheet 44** Mon 1st Mar. 2027 *Table of variations; maximum and minimum on an interval*

1. Warm-up · Quick Calculate $x^2$ for $x = -2$, 0, 3.
2. Core · Method $f$ is decreasing on $[-3, 1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1, 4]$ with $f(4) = 4$. Give the maximum and the minimum of $f$ on $[-3, 4]$.
3. Core + · True/false True or false? Justify. The maximum of a function on $[a, b]$ is always reached at $a$ or at $b$.
4. Review · Table Is this table a proportionality table?

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $y$ | 5 | 9 | 13 |

5. Thread · Method Is the triangle with sides 5, 12 and 13 right-angled? Which property should be used?

**Sheet 45** Tue 2 Mar. 2027 *Affine functions: rate of change and variations*

1. Warm-up · Quick $f(x) = 3x - 1$. Calculate $\dfrac{f(4) - f(1)}{4 - 1}$.
2. Core · Quick Direction of variation of $f(x) = -4x + 7$?
3. Core + · Proof Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.
4. Review · Table Ages of the members of a club.

| Age | 10 | 11 | 12 | 13 |
|---|---|---|---|---|
| Frequency | 3 | 6 | 10 | 6 |

   How many members are at most 11 years old? What relative frequency is that?
5. Thread · Programme → formula Programme: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.

**Sheet 46** Thu 4 Mar. 2027 *Absolute value and square functions: sign and variations*

1. Warm-up · Quick Calculate $|-5|$, $(-5)^2$ and $|2-7|$.
2. Core · Method Give the table of variations of $x \mapsto |x|$ and its sign.
3. Core + · Reasoning Compare $(-3)^2$ and $(-2)^2$ then $3^2$ and $2^2$. Is the square function increasing on $\mathbb{R}$?
4. Review · Method Two fair dice are rolled. Probability that the sum is 7?
5. Thread · Method Programme: choose a number; add 1; square it; subtract the square of the starting number. What is the result for 6? What is the general expression?

**Sheet 47** Mon 8 Mar. 2027 *Reciprocal function: variations; relative position of $y = x$ and $y = x^2$*

1. Warm-up · Quick Calculate $\dfrac{1}{2}$, $\dfrac{1}{-4}$ and $\dfrac{1}{0.1}$.
2. Core · Method Compare $\dfrac{1}{3}$ and $\dfrac{1}{5}$; what can be said about the variations of the reciprocal function on $(0, +\infty)$?
3. Core + · True/false True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.
4. Review · Quick What are the front view and the top view of a cylinder standing on its base?
5. Thread · Prove Show that the product of two even integers is a multiple of 4.

**Sheet 48** Tue 9 Mar. 2027 *Comparing $f(a)$ and $f(b)$; optimisation problems*

1. Warm-up · Quick $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.
2. Core · Method $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculating, using the variations.
3. Core + · Optimisation $A(x) = x(10 - x)$ for $x \in [0, 10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x-5)^2$ and deduce the maximum.
4. Review · Quick 18 pupils out of 24 have an average above 10. Give the relative frequency as a percentage.
5. Thread · MCQ The solution of $\dfrac{x}{3} + 1 = \dfrac{7}{2}$ is:

   **A.** $\dfrac{15}{2}$ **B.** $\dfrac{9}{2}$ **C.** $\dfrac{21}{2}$ **D.** $\dfrac{5}{6}$